

專案名稱： 企業級多協定路 (Multi-Routing Protocols) 由重分發 (Redistribution) 與 GRE over IPsec 安全 VPN 整合實作

平台： Cisco IOS / GNS3

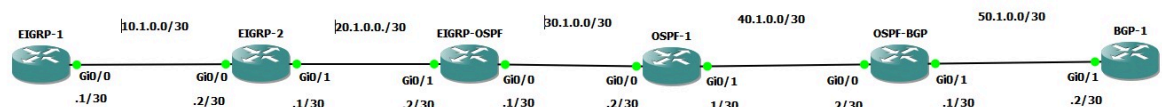
關鍵技術： Multi-Routing Protocols (EIGRP, OSPF, BGP), Route Redistribution, GRE over IPsec VPN

傳統問題： 異構網路 (多路由協定) 難以互通，且跨公網傳輸缺乏安全加密與動態路由支援

- **高可用性與擴充性 (Availability & Scalability)：** 企業包含多個自治系統與區域 (EIGRP AS 100, OSPF Area 0, BGP AS 64512/64513)，傳統點對點靜態設定擴充困難。本專案透過邊界路由器雙向路由重分發 (Route Redistribution) 實現全網動態互通，並搭配 GRE 隧道架構封裝動態路由協定封包。
- **安全性：** 純 GRE 隧道缺乏加密能力 (明文傳輸)，為了保護敏感企業資料，疊加 **IPsec (IKEv1/ESP)** 進行機密性保護。同時關閉路由器與交換器上未使用的協定及遺留功能 (如關閉未使用的介面) 以增加安全性與防範網路攻擊

Logical Topology

Project - EIGRP/OSPF/BGP Redistribution + GRE over IPsec



IP Address Table

EIGRP-1

Interface	IP address	Subnet Mask	Description
g0/0	10.1.0.1	/30	connect to router EIGRP-2
g0/1-7			shut down
Tunnel253	192.168.100.1	/30	tunnel to router BGP-1

EIGRP-2

Interface	IP address	Subnet Mask	Description
g0/0	10.1.0.2	/30	connect to router EIGRP-1
g0/1	20.1.0.1	/30	connect to router EIGRP-OSPF
g0/2-3			shut down

EIGRP-2

Interface	IP address	Subnet Mask	Description
g0/0	30.1.0.1	/30	connect to router OSPF-1
g0/1	20.1.0.2	/30	connect to router EIGRP-2
g0/2-3			shut down

OSPF-1

Interface	IP address	Subnet Mask	Description
g0/0	30.1.0.2	/30	connect to router EIGRP-OSPF
g0/1	40.1.0.1	/30	connect to router OSPF-BGP
g0/2-3			shut down

OSPF-BGP

Interface	IP address	Subnet Mask	Description
g0/0	40.1.0.2	/30	connect to router OSPF-1
g0/1	50.1.0.1	/30	connect to router BGP-1
g0/2-3			shut down

BGP-1

Interface	IP address	Subnet Mask	Description
g0/0			shut down
g0/1	50.1.0.2	/30	connect to router OSPF-BGP
g0/2			shut down
Tunnel253	192.168.100.2	/30	tunnel to router EIGRP-1

EIGRP Neighborhood

```
EIGRP-2#sh ip eigrp neighbors
EIGRP-IPv4 VR(EIGRP_1) Address-Family Neighbors for AS(100)
H   Address                Interface                Hold Uptime    SRTT    RTO    Q    Seq
                                (sec)              (ms)          Cnt  Num
1   20.1.0.2                Gi0/1                  12 00:04:02    1    100    0    11
0   10.1.0.1                Gi0/0                  14 00:19:24    1    100    0    11
```

OSPF Neighborhood

```
OSPF-1#sh ip ospf ne
OSPF-1#sh ip ospf neighbor

Neighbor ID    Pri   State           Dead Time   Address      Interface
50.1.0.1       1     FULL/BDR        00:00:33    40.1.0.2     GigabitEthernet0/1
30.1.0.1       1     FULL/DR         00:00:36    30.1.0.1     GigabitEthernet0/0
```

BGP Neighborhood

```
OSPF-1#sh ip ospf ne
OSPF-1#sh ip ospf neighbor

Neighbor ID    Pri   State           Dead Time   Address      Interface
50.1.0.1       1     FULL/BDR        00:00:33    40.1.0.2     GigabitEthernet0/1
30.1.0.1       1     FULL/DR         00:00:36    30.1.0.1     GigabitEthernet0/0
```

Router EIGRP-1 traceroute to Router BGP-2

```
EIGRP-1#sh ip ro
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.1.0.0/30 is directly connected, GigabitEthernet0/0
L       10.1.0.1/32 is directly connected, GigabitEthernet0/0
D       20.0.0.0/16 is subnetted, 1 subnets
D       20.1.0.0 [90/15360] via 10.1.0.2, 02:06:55, GigabitEthernet0/0
D       30.0.0.0/30 is subnetted, 1 subnets
D EX    30.1.0.0 [170/1034240] via 10.1.0.2, 01:51:33, GigabitEthernet0/0
D       40.0.0.0/30 is subnetted, 1 subnets
D EX    40.1.0.0 [170/1034240] via 10.1.0.2, 01:50:27, GigabitEthernet0/0
D       50.0.0.0/30 is subnetted, 1 subnets
D EX    50.1.0.0 [170/1034240] via 10.1.0.2, 01:38:07, GigabitEthernet0/0
C       192.168.100.0/24 is variably subnetted, 2 subnets, 2 masks
L       192.168.100.0/30 is directly connected, Tunnel253
L       192.168.100.1/32 is directly connected, Tunnel253
EIGRP-1#
EIGRP-1#
EIGRP-1#trac
EIGRP-1#traceroute 50.1.0.2
Type escape sequence to abort.
Tracing the route to 50.1.0.2
VRF info: (vrf in name/id, vrf out name/id)
  1 10.1.0.2 3 msec 2 msec 1 msec
  2 20.1.0.2 4 msec 4 msec 4 msec
  3 30.1.0.2 4 msec 4 msec 4 msec
  4 40.1.0.2 5 msec 5 msec 4 msec
  5 50.1.0.2 6 msec 5 msec 5 msec
EIGRP-1#
```

Router BGP-2 ping to EIGRP-1 through the tunnel 253 (GRE over IPsec)

```
BGP-1#sh ip int br
Interface                               IP-Address      OK? Method Status          Protocol
GigabitEthernet0/0                     unassigned      YES NVRAM    administratively down down
GigabitEthernet0/1                     50.1.0.2        YES manual    up              up
GigabitEthernet0/2                     unassigned      YES NVRAM    administratively down down
GigabitEthernet0/3                     unassigned      YES NVRAM    administratively down down
NVI0                                    50.1.0.2        YES unset    up              up
Tunnel253                              192.168.100.2   YES manual    up              up
BGP-1#
BGP-1#
BGP-1#
BGP-1#ping 192.168.100.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.100.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 9/10/14 ms
```

IPsec Phase_1 (show crypto isakmp sa)

```
BGP-1#show crypto isakmp sa
IPv4 Crypto ISAKMP SA
dst          src          state         conn-id status
10.1.0.1     50.1.0.2     QM_IDLE       1001 ACTIVE
```

IPsec Phase_2 (show crypto ipsec sa)

```
EIGRP-1#[A
% Bad IP address or host name
% Unknown command or computer name, or unable to find computer address
EIGRP-1#show crypto ipsec sa

interface: Tunnel253
  Crypto map tag: Tunnel253-head-0, local addr 10.1.0.1

  protected vrf: (none)
  local ident (addr/mask/prot/port): (10.1.0.1/255.255.255.255/47/0)
  remote ident (addr/mask/prot/port): (50.1.0.2/255.255.255.255/47/0)
  current_peer 50.1.0.2 port 500
    PERMIT, flags={origin is acl,}
    #pkts encaps: 10, #pkts encrypt: 10, #pkts digest: 10
    #pkts decaps: 10, #pkts decrypt: 10, #pkts verify: 10
    #pkts compressed: 0, #pkts decompressed: 0
    #pkts not compressed: 0, #pkts compr. failed: 0
    #pkts not decompressed: 0, #pkts decompress failed: 0
    #send errors 0, #recv errors 0
```

Shut down non-used port

```
EIGRP-1#sh ip int br
Interface                               IP-Address      OK? Method Status          Protocol
GigabitEthernet0/0                      10.1.0.1        YES manual up              up
GigabitEthernet0/1                      unassigned      YES NVRAM   administratively down down
GigabitEthernet0/2                      unassigned      YES NVRAM   administratively down down
GigabitEthernet0/3                      unassigned      YES NVRAM   administratively down down
GigabitEthernet0/4                      unassigned      YES NVRAM   administratively down down
GigabitEthernet0/5                      unassigned      YES NVRAM   administratively down down
GigabitEthernet0/6                      unassigned      YES NVRAM   administratively down down
GigabitEthernet0/7                      unassigned      YES NVRAM   administratively down down
NVI0                                     10.1.0.1        YES unset  up              up
Tunnel253                               192.168.100.1   YES manual up              up
```

Only SSH Remote connection allowed & unclear password

```
line vty 5 15
 password 7 047828283F
 login
 transport input ssh
,
```